

Operation Procedure:

Connect the Charger to Battery first using the Battery Connector cable. Apply mains power using a 6A rated mains socket. The charger will blink all five indications once and a characteristic beep sound at power-up, the Green mains indication will become steady and the charger will start charging. The Battery Bar Indications will start animating up-down with current increasing gradually. Ensure the charger is kept in a well-ventilated area and the ventilation slots on the charger must be un-obstructed for air flow. As the charging current increases, the charger turns on the internal cooling fan and turns off the fan once the current goes down or internal temperature falls. Once the Battery voltage reaches set voltage and current falls to a pre-defined level, the Battery Bar Indications becomes steady. The battery is almost full charged at this instant and can be disconnected and used for backup. However, the charger continues to charge the battery for some more time till the battery draws less than 5% of the CC current. Once the battery is fully charged the charger stops and gives a characteristic beep sound will the Battery Bar indication steady on.

LED indications:

Indication	Condition
Mains Led Steady on	Charging On from Mains
Mains Led Blink	Mains out of Acceptable range.
Battery Bar All Four Led Steady on	Charging Complete
Battery Bar One or More Led Up-Down	Charging On
One or More Battery Bar Led Steady on	Battery Voltage Level showing – Not Charging
1 Battery Bar Led 25 Blink	Deep Discharged Battery
2 Battery Bar LED 100,25 Blink	Fault Indication occurring immediately after turning on power.
	Battery Reverse Connection OR Output Short OR Battery Voltage too High.
	Battery Connection Open Circuit OR PCM Disconnect in already full charged battery.
	PCM Disconnect due to battery full charge.
2 Battery Bar LED 100, 75 Blink	Over Temperature Fault
2 Battery Bar LED 50, 25 Blink	Internal Fault–No output Current
All Four Battery Bar LED Blink	Internal Fault –Over Current

Mfg. & Mktd. by :

ECOSTAR INNOVATION PVT. LTD.

Khasara No-218, Nangli Sakrawati Industrial Area, Najafgarh, New Delhi-110043
E-mail : info@ecostartek.com | Mobile : +91 9818511898



Ecostar®

E-Vehicle
Lithium Battery Charger
400W



DESIGNED & MADE
INDIA

This product is a SMPS type 2-Wheeler E-Vehicle battery charger optimized for charging various Lithium chemistry like LFP, NCM or NCA battery packs. Various models are available in this series for charging 13-Cell-Series to 19-Cell-Series Battery packs.

Common Features

- charging Current and Voltage profile optimized for Lithium LFP or NCM or NCA chemistry battery packs for 2 Wheeler E-Vehicle application.
- Configurable for various charging voltages in the range of 53V to 72V.
- 400W Power Rating – Charging current range 5A to 7A depending on voltage setting.
- High Efficiency to reduce Power Consumption from Mains.
- Electronic Reverse Battery or Output Short Protection – enhanced safety.
- Informative Battery Bar Graph and Mains indication along with audio alarms.
- Rugged metal enclosure.
- Easy to handle due to light weight

Model List :

Models – Low Voltage	ES5407	ES5707	ES5806
Battery Pack Type	15S-LFP 13S-NCM	15S-LFP 13S-NCM	16S-LFP
Nominal Battery Voltage	48V	48V	51V
Max. Charging Voltage	54V	57V	58.0V
Max. Charging Current	7A	7A	6A
Models – High Voltage	ES6706	ES6906	ES7106
Battery Pack Type	19S-LFP 16S-NCM	19S-LFP 20S-NCM	20S-LFP 17S-NCM
Nominal Battery Voltage	60V	60V	60V
Max. Charging Voltage	67.0V	69.0V	71.0V
Max. Charging Current	6A	6A	6A

* Other Models are also available based on requirement.

Specifications

Max Output Power	400W MAX
CV Voltage Range	Factory preset (Max. 71V)
CC Current Range	Factory preset (Max 7A)
CC Current Tolerance	CC Setting $\pm$ 0.5A
CV Voltage Tolerance	CV Setting $\pm$ 0.3V
Efficiency	91% Typical
Mains Opreating Range	150VAC -280VAC $\pm$ 10V
Mains Opreating Frequency	40 Hz TO 60 Hz
Max Input Current	2A (Output current de-rated at low mains to limit max Input current)
Cooling System	Forced Air Cooled
Visual Indication	LED Based
Audio Indications (Buzzer)	Charger Startup, Battery Charge Completion – Characteristic Pattern Beep. Mains Failure or any Fault – Single Beeps
Protection	Reverse Battery (Electronically) Mains High Voltage (Withstand up to 320VAC RMS.), Mains Over Current (By Fuse) Thermal Protection (Electronically)
Charger Operating Temprature	0°C TO 45°C
Humidity	95% RH Non-Condensing
Battery Cable	1.3m, 1.5sqmm, Insulated 2 Core Cable with IEC C13 Connector
Mains Cable	1.5m, 0.50sqmm 3core ISI Marked Cable With 6A Plug
Enclosure	Powder coated sheet metal cabinet
Dimension / Net Weight	200mm x 135mm x 70mm / 1.7 kg

* Specification can be changed by company without any prior notice for regular up gradations.